

MAJD ALKAYYAL

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Senior Software Engineer | LLM & Agent Systems | NLP | RAG | Distributed Systems

LLM-focused software engineer with experience building multi-agent systems, RAG pipelines, and production AI features. Published research (ECML 2025) on agent-based decision systems outperforming GPT-4o in structured reasoning tasks. Strong background in distributed systems, backend architecture, and scalable AI infrastructure.

SKILLS

Programming Languages	C#, C++, Python, TypeScript
Specialty Areas	LLMs, RAG, Multi-Agent Systems, Databases, Backend Systems
Frameworks & Technologies	.NET Core, FastAPI, NestJS, Flutter, NextJS, Postgres, Mongo, LangChain, LangGraph, Ray, Codex
Infra	AWS, Azure, Docker, CI/CD

WORK EXPERIENCE

KPMG Deutschland Senior Software Engineer - Assistant Manager	Munich, Germany Jul. 2024 - Present
• Co-led a global engineering team (25+), delivering a large-scale Transfer Pricing platform and driving feature development, system design, and release management • Worked directly with clients and stakeholders to translate business requirements into scalable systems, aligning technical solutions with domain-specific needs • Designed and implemented AI-assisted features for document-heavy workflows, including LLM-based content extraction and report drafting support • Built a document processing pipeline using LLMs to extract structured information from financial/legal documents and automate report sections • Developing a context-aware AI assistant (early-stage) that operates within user sessions, guiding workflows and suggesting next actions based on system state and user context.	
PwC Deutschland Associate Software Engineer (Working Student)	Munich, Germany Apr. 2023 – Jun. 2024
• Built and maintained PwC’s internal AuthaaS platform, providing unified SSO for multiple internal applications. • Implemented version-controlled decision-tree backends for automated workflow systems to improve auditability. • Developed and executed integration tests to detect regressions in legacy services, reducing post-release incidents. • Improved observability by integrating Prometheus and creating Grafana dashboards for key service metrics. • Authored detailed API and backend documentation to streamline onboarding and handovers.	
TelyPay (Maaly) Senior Software Engineer, Project Manager & Technical Consultant	Muscat, Oman Feb. 2022 – Nov. 2024
• Rebuilt the company’s fintech platform into a unified, scalable architecture, improving stability and performance • Led a team of 6 engineers, driving system design, code quality, and delivery • Designed and deployed a financial insights platform on AWS, enabling real-time analytics over aggregated banking data • Built an LLM-powered chatbot and insights dashboard, allowing natural language querying of financial data • Implemented an agent workflow with tool use, enabling the LLM to query databases and retrieve structured insights • Designed a retrieval + context construction pipeline (RAG-style) and a hybrid LLM + rule-based system for reliable transaction classification and insights	

- Designed and deployed the Dar Al Atta’a Charity App, enabling mobile and web donations for local causes.
- Integrated online payments, SMS, and email notifications, improving the efficiency of donation and case management.
- Hosted and managed infrastructure on Oman Data Park IaaS, ensuring secure and stable operations.
- Delivered the project independently from concept to deployment, marking the company’s first fully in-house mobile product.

RESEARCH EXPERIENCE

McKinsey & Company / Technical University of Munich (TUM)
LLM Research Engineer – StrategicAI Project (Master’s Thesis)

Munich, Germany
Oct. 2024 – Apr. 2025

- Built a multi-agent LLM system with autonomous role-based collaboration, tool use, and memory
- Designed RAG pipelines with dynamic retrieval + grounding for decision reasoning
- Developed agent orchestration framework (task decomposition, planning, tool routing)
- Achieved performance surpassing GPT-4o on structured decision benchmarks (ECML 2025)

Fusionbase / Technical University of Munich (TUM)
AI Systems Engineer – Project Lead (Industry-Academia Collaboration)

Munich, Germany
Oct. 2023 – Jun. 2024

- Led a 4-member team to design and build CreateData4AI, a collaborative annotation platform for data-centric AI.
- Developed the full-stack web platform (Next.js + FastAPI + MongoDB) enabling distributed labeling.
- Engineered the GPU/CPU-optimized pipeline using Ray for distributed embedding computation.
- Implemented real-time collaboration features, enabling multiple users to jointly perform annotation refinement tasks.

TEACHING EXPERIENCE

Technical University of Munich (TUM)
Teaching Assistant — Foundations in Data Engineering

Munich, Germany
Oct. 2023 – Feb. 2024

- Taught C++, SQL, distributed consensus algorithms (Raft), and knowledge graphs to Master’s students.
- Supported course instruction with tutorials, problem-solving sessions, and guidance on advanced data engineering concepts.

German University of Technology in Oman (GUtech)
Teaching Assistant — C++ / Data Structures

Muscat, Oman
Jun. 2021 – Apr. 2022

- Mentored first- and second-year undergraduate students in foundational programming and data structures.
- Led lab sessions, graded submissions, and provided hands-on support to improve student understanding.

EDUCATION

Technical University of Munich (TUM)
M.Sc. Data Engineering and Analytics

Munich, Germany
Oct. 2022 – Apr. 2025

- cGPA 1.3/1.0
- Graduated in the top 10% of the cohort; recipient of the DAAD Master Scholarship for academic excellence.
- Served as Teaching Assistant for *Foundations in Data Engineering*, teaching C++, SQL, distributed consensus algorithms (Raft), and knowledge graphs to Master’s students.
- Relevant coursework: Natural Language Processing (NLP), Advanced NLP, Foundations in Data Analysis, Introduction to Deep Learning, Distributed Systems.

German University of Technology in Oman (GUtech)
B.Sc. Computer Science

Muscat, Oman
Jun. 2016 – Sep. 2021

- Graduated with High Distinction as the top-ranked student in the university, achieving a perfect 4.0/4.0 GPA.
- Graduated first in the university, awarded the Academic Rector’s Medal and a DAAD Bachelor Thesis Scholarship for research at RWTH Aachen.
- Bachelor Thesis (RWTH Aachen): Designed and implemented an open educational resource mobile app (Flutter + .NET Core) adopted by German schools to teach Creative Commons licensing; achieved 80%+ usability satisfaction in pilot studies.
- Served as a Teaching Assistant for C++/Data Structures for one academic year, mentoring first/second-year students.

HONORS & AWARDS

DAAD Master Scholarship

Awarded for top 5% academic excellence and research potential.

Munich, Germany
Oct. 2022 – Jul. 2024

Academic Rector's Medal

Conferred to the top graduating student across all majors at GUtech.

Muscat, Oman
Mar. 2022

DAAD Bachelor Thesis Scholarship

Secured one of two GUtech–RWTH scholarship seats to conduct thesis research in Germany on Open Educational Resources.

Aachen, Germany
Apr. 2021 – Sep. 2021

Oman Education Services (OES) Scholarship

Full merit-based scholarship; first and only recipient in the university's history that year.

Muscat, Oman
Jun. 2016 – Sep. 2021

ICPC Oman Champion – International Collegiate Programming Contest

First place nationally, qualified for the Egypt regional.

Muscat, Oman
2019

LANGUAGES

English – C1

German – A2

Arabic – Native